

5.1 Motivation and Problem Setup

Consider the equation $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ is an $m \times n$ matrix, $\mathbf{x} \in \mathbb{R}^n$, and $\mathbf{b} \in \mathbb{R}^m$. We know that this system has a solution if and only if $\mathbf{b}$ lies in the column space of $\mathbf{A}$, i.e., $\mathbf{b} \in \text{Col}(\mathbf{A})$.

But what happens when $\mathbf{b} \notin \text{Col}(\mathbf{A})$? This situation arises frequently in practice, especially when the system is **over-determined**, which occurs when there are more equations than unknowns ($m > n$). In such cases, we generally cannot satisfy all m equations simultaneously, and no exact solution exists.

As a compromise, we look for the vector $\hat{\mathbf{x}}$ that comes as close as possible to satisfying $\mathbf{Ax} = \mathbf{b}$. The **residual** $\mathbf{r} = \mathbf{b} - \mathbf{Ax}$ measures how far off we are from an exact solution, and we quantify “closeness” by the squared Euclidean norm of the residual. This leads to the **least squares problem**:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|^2 \quad (1)$$

The quantity $\|\mathbf{Ax} - \mathbf{b}\|^2 = r_1^2 + r_2^2 + \dots + r_m^2$ is the sum of the squared residuals across all m equations, which is why this method is called *least squares*.

5.2 Orthogonal Projection

The key geometric idea behind least squares is **orthogonal projection**. We start with the simplest case of projecting onto a vector and then generalize to projecting onto a subspace.

5.2.1 Projection onto a Single Vector

Given vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ with $\mathbf{a} \neq \mathbf{0}$, we want to find the scalar $\hat{x} \in \mathbb{R}$ such that $\hat{x}\mathbf{a}$ is the point on the line spanned by $\mathbf{a}$ that is closest to $\mathbf{b}$. In other words, we want to minimize $\|\mathbf{b} - x\mathbf{a}\|^2$ over all scalars x .

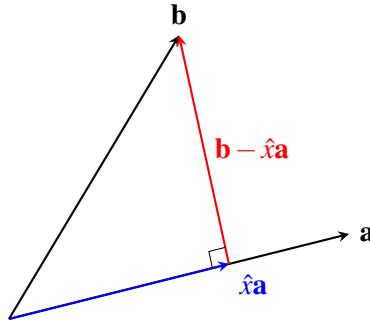


Figure 1: The projection $\hat{x}\mathbf{a}$ is the closest point on the line through $\mathbf{a}$ to the vector $\mathbf{b}$.

The closest point is achieved when the residual $\mathbf{b} - \hat{x}\mathbf{a}$ is orthogonal to $\mathbf{a}$. Otherwise, if the residual had any component along $\mathbf{a}$, we could achieve a smaller error (squared Euclidean norm of the residual) by moving along $\mathbf{a}$.

Setting the inner product of $\mathbf{a}$ and $\mathbf{b} - \hat{x}\mathbf{a}$ to zero due to orthogonality:

$$\begin{aligned}\langle \mathbf{a}, \mathbf{b} - \hat{x}\mathbf{a} \rangle &= 0 \\ \langle \mathbf{a}, \mathbf{b} \rangle - \hat{x}\langle \mathbf{a}, \mathbf{a} \rangle &= 0 \\ \hat{x} &= \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\langle \mathbf{a}, \mathbf{a} \rangle} = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{a}^\top \mathbf{a}}\end{aligned}$$

The projected vector is:

$$\text{proj}_{\mathbf{a}}(\mathbf{b}) = \hat{x}\mathbf{a} = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{a}^\top \mathbf{a}} \mathbf{a} \quad (2)$$

We can also write this as a matrix transformation on $\mathbf{b}$:

$$\text{proj}_{\mathbf{a}}(\mathbf{b}) = \frac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top \mathbf{a}} \mathbf{b} = \mathbf{P}_{\mathbf{a}} \mathbf{b} \quad (3)$$

where $\mathbf{P}_{\mathbf{a}} = \frac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top \mathbf{a}}$ is the **projection matrix** onto the span of $\mathbf{a}$. Note that $\mathbf{a}\mathbf{a}^\top$ is an outer product (an $n \times n$ matrix), while $\mathbf{a}^\top \mathbf{a}$ is a scalar.

5.2.2 Projection onto a Subspace

Now suppose we want to project $\mathbf{b}$ not onto a single vector, but onto the column space of a matrix $\mathbf{A}$ with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. We seek the vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ such that $\mathbf{A}\hat{\mathbf{x}}$ is the point in $\text{Col}(\mathbf{A})$ closest to $\mathbf{b}$.

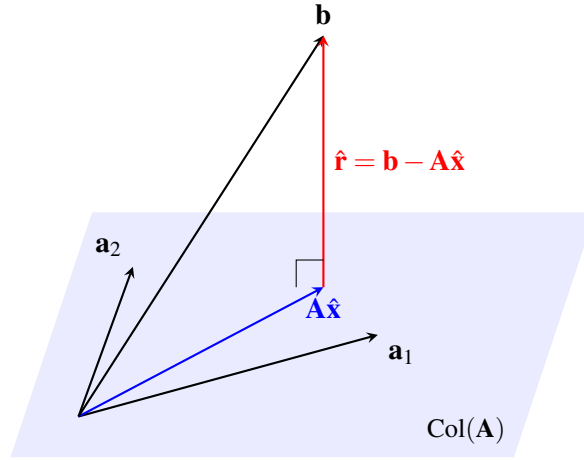


Figure 2: For the least squares solution $\hat{\mathbf{x}}$, the vector $\mathbf{A}\hat{\mathbf{x}}$ is the projection of $\mathbf{b}$ onto $\text{Col}(\mathbf{A})$. The residual $\hat{\mathbf{r}}$ is orthogonal to $\text{Col}(\mathbf{A})$.

The same geometric principle applies: the residual $\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$ must be orthogonal to the column space of $\mathbf{A}$. Since the columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ span $\text{Col}(\mathbf{A})$, it suffices to require orthogonality to each column:

$$\mathbf{a}_j^\top (\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) = 0 \quad \text{for } j = 1, \dots, n \quad (4)$$

Stacking these n conditions into a single matrix equation gives:

$$\mathbf{A}^\top (\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) = \mathbf{0} \quad (5)$$

Rearranging, we obtain the **normal equations**, the fundamental equations for least squares:

$$\boxed{\mathbf{A}^\top \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^\top \mathbf{b}} \quad (6)$$

5.3 The Least Squares Solution

5.3.1 Solving the Normal Equations

The matrix $\mathbf{A}^\top \mathbf{A}$ is the **Gram matrix** of $\mathbf{A}$. A key property of the Gram matrix is that if the columns of $\mathbf{A}$ are linearly independent, then $\mathbf{A}^\top \mathbf{A}$ is invertible.

Proof. Assume the columns of $\mathbf{A}$ are linearly independent. Then $\mathbf{A}$ has a trivial null space, i.e., the only solution to

$\mathbf{Ax} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$. Now consider the equation $\mathbf{A}^\top \mathbf{Ax} = \mathbf{0}$:

$$\mathbf{A}^\top \mathbf{Ax} = \mathbf{0} \implies \mathbf{x}^\top \mathbf{A}^\top \mathbf{Ax} = 0 \implies (\mathbf{Ax})^\top (\mathbf{Ax}) = 0 \implies \|\mathbf{Ax}\|^2 = 0 \implies \mathbf{Ax} = \mathbf{0}$$

So any $\mathbf{x}$ in $N(\mathbf{A}^\top \mathbf{A})$ is also in $N(\mathbf{A})$, implying that $N(\mathbf{A}^\top \mathbf{A}) \subseteq N(\mathbf{A})$. The reverse inclusion $N(\mathbf{A}) \subseteq N(\mathbf{A}^\top \mathbf{A})$ can also be shown: if $\mathbf{Ax} = \mathbf{0}$, then $\mathbf{A}^\top \mathbf{Ax} = \mathbf{A}^\top \mathbf{0} = \mathbf{0}$. Therefore $N(\mathbf{A}^\top \mathbf{A}) = N(\mathbf{A})$. Since $\mathbf{A}$ has a trivial null space, so does $\mathbf{A}^\top \mathbf{A}$, and since $\mathbf{A}^\top \mathbf{A}$ is an $n \times n$ square matrix, it is invertible. $\square$

Throughout this note, we assume the columns of $\mathbf{A}$ are linearly independent unless stated otherwise. Under this assumption, the normal equations have a unique solution:

$$\boxed{\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}} \quad (7)$$

This is the **least squares solution** of $\mathbf{Ax} = \mathbf{b}$. Appendix A describes another way to compute it, using the QR factorization.

5.3.2 Proving Optimality of the Least Squares Solution

We now show that $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$ uniquely minimizes $\|\mathbf{Ax} - \mathbf{b}\|^2$, where $\mathbf{A}$ is an $m \times n$ matrix with linearly independent columns and $\mathbf{b} \in \mathbb{R}^m$.

Proof. For any $\mathbf{x} \in \mathbb{R}^n$, we split $\mathbf{Ax} - \mathbf{b}$ into two parts:

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = \|\underbrace{(\mathbf{Ax} - \mathbf{A}\hat{\mathbf{x}})}_{\mathbf{u}} + \underbrace{(\mathbf{A}\hat{\mathbf{x}} - \mathbf{b})}_{\mathbf{v}}\|^2$$

Expanding the squared norm using $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\mathbf{u}^\top \mathbf{v}$:

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = \|\mathbf{Ax} - \mathbf{A}\hat{\mathbf{x}}\|^2 + \|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 + 2(\mathbf{Ax} - \mathbf{A}\hat{\mathbf{x}})^\top (\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) \quad (8)$$

The cross term disappears due to orthogonality:

$$\begin{aligned} (\mathbf{Ax} - \mathbf{A}\hat{\mathbf{x}})^\top (\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) &= (\mathbf{x} - \hat{\mathbf{x}})^\top \mathbf{A}^\top (\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) \\ &= (\mathbf{x} - \hat{\mathbf{x}})^\top (\mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}} - \mathbf{A}^\top \mathbf{b}) \\ &= (\mathbf{x} - \hat{\mathbf{x}})^\top \mathbf{0} = 0 \end{aligned}$$

where we used the normal equations $\mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^\top \mathbf{b}$ in the last step.

Therefore, (8) simplifies to:

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = \|\mathbf{A}(\mathbf{x} - \hat{\mathbf{x}})\|^2 + \|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 \geq \|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 \quad (9)$$

This shows that $\hat{\mathbf{x}}$ is a minimizer. For **uniqueness**: suppose $\mathbf{x} \neq \hat{\mathbf{x}}$. Then $\mathbf{x} - \hat{\mathbf{x}} \neq \mathbf{0}$, and since $\mathbf{A}$ has linearly independent columns, $\mathbf{A}(\mathbf{x} - \hat{\mathbf{x}}) \neq \mathbf{0}$, so $\|\mathbf{A}(\mathbf{x} - \hat{\mathbf{x}})\|^2 > 0$. Therefore $\|\mathbf{Ax} - \mathbf{b}\|^2 > \|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2$, meaning no other vector achieves the same minimum. Thus $\hat{\mathbf{x}}$ is the unique minimizer. $\square$

5.4 Projection Matrices

We have seen that $\mathbf{A}\hat{\mathbf{x}}$ is the point in $\text{Col}(\mathbf{A})$ closest to $\mathbf{b}$. The map $\mathbf{b} \mapsto \mathbf{A}\hat{\mathbf{x}}$ is therefore a **projection** onto $\text{Col}(\mathbf{A})$. Substituting the least squares formula:

$$\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} = \mathbf{Pb} \quad (10)$$

where

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \quad (11)$$

is the **projection matrix** onto $\text{Col}(\mathbf{A})$.

Definition 5.1 (Projection Matrix): A matrix $\mathbf{P}$ is an **orthogonal projection matrix** if it satisfies:

1. **Idempotence:** $\mathbf{P}^2 = \mathbf{P}$. Projecting twice is the same as projecting once. That is, if a vector is already in the subspace, projecting it again does nothing.
2. **Symmetry:** $\mathbf{P}^\top = \mathbf{P}$.

We verify both properties for $\mathbf{P} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$:

Idempotence:

$$\begin{aligned} \mathbf{P}^2 &= \left[\mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \right] \left[\mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \right] \\ &= \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \underbrace{(\mathbf{A}^\top \mathbf{A})(\mathbf{A}^\top \mathbf{A})^{-1}}_{=\mathbf{I}_n} \mathbf{A}^\top \\ &= \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top = \mathbf{P} \end{aligned}$$

Symmetry:

$$\begin{aligned} \mathbf{P}^\top &= \left[\mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \right]^\top \\ &= \mathbf{A} \left[(\mathbf{A}^\top \mathbf{A})^{-1} \right]^\top \mathbf{A}^\top \\ &= \mathbf{A} \left[(\mathbf{A}^\top \mathbf{A})^\top \right]^{-1} \mathbf{A}^\top \\ &= \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top = \mathbf{P} \end{aligned}$$

5.5 Application: Linear Regression and Curve Fitting

5.5.1 Linear Regression

One of the most common applications of least squares is **linear regression**. Suppose we observe M data points $(t_1, y_1), (t_2, y_2), \dots, (t_M, y_M)$ and wish to find the line

$$y(t) = \alpha + \beta t$$

that best fits the data in the least squares sense.

Each data point gives one equation $\alpha + \beta t_k = y_k$. Stacking all M equations:

$$\underbrace{\begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_M \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \alpha \\ \beta \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_M \end{bmatrix}}_{\mathbf{y}} \quad (12)$$

When $M > 2$, this system is generally over-determined, and the least squares solution $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{y}$ gives the best-fit line in terms of minimizing the total squared error $\sum_{k=1}^M (\alpha + \beta t_k - y_k)^2$.

Example 5.1 (Linear regression):

Suppose we measure the position of a moving object at times $t_1 = 0, t_2 = 1, t_3 = 2$ and observe positions $y(0) = 1, y(1) = 3, y(2) = 5$.

The system is:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

We compute the components of the normal equations:

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \\ \mathbf{A}^\top \mathbf{y} &= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 9 \\ 13 \end{bmatrix} \end{aligned}$$

Inverting the Gram matrix:

$$(\mathbf{A}^\top \mathbf{A})^{-1} = \frac{1}{15-9} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}$$

Computing the least squares solution:

$$\hat{\mathbf{x}} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 13 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 45-39 \\ -27+39 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 6 \\ 12 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

So $\alpha = 1$ and $\beta = 2$, giving the best-fit line $y(t) = 1 + 2t$.

We can verify: $y(0) = 1$, $y(1) = 3$, $y(2) = 5$. In this case the data is perfectly collinear, so the least squares solution exactly fits all three points and the residual is $\mathbf{0}$.

In Appendix A, we solve this same system using QR factorization and obtain the same result.

5.5.2 Curve Fitting

The least squares framework is not limited to fitting lines. We can fit any model that is linear in the unknown parameters, even if it is nonlinear in the input variables.

For example, to fit a circle $a(x^2 + y^2) + dx + ey = 1$ to data points $(x_1, y_1), \dots, (x_M, y_M)$, we set up:

$$\underbrace{\begin{bmatrix} x_1^2 + y_1^2 & x_1 & y_1 \\ x_2^2 + y_2^2 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ x_M^2 + y_M^2 & x_M & y_M \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} a \\ d \\ e \end{bmatrix}}_{\mathbf{p}} = \underbrace{\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}}_{\mathbf{b}} \quad (13)$$

Even though the equations involve nonlinear terms like x^2 and y^2 , these are computed from the known data and simply become entries of the matrix $\mathbf{A}$. The unknowns enter linearly, which allows us to use least squares. (We call the vector of unknowns $\mathbf{p}$ here, rather than $\mathbf{x}$, to avoid confusion with the data coordinates $x_1, \dots, x_M$.)

A Appendix: Solving Least Squares via QR Factorization

(Note: QR is not a large topic of emphasis in the course, though we expect students to be familiar with the concept. QR has many deep connections to least squares.)

The formula $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$ gives us the least squares solution directly, but computing it requires forming the product $\mathbf{A}^\top \mathbf{A}$ and then solving the normal equations. In practice, this can be numerically unstable: when we multiply $\mathbf{A}^\top \mathbf{A}$, small errors in the entries of $\mathbf{A}$ (e.g., from measurement noise or the finite precision of computer arithmetic) can become amplified.

QR factorization provides an alternative that avoids forming $\mathbf{A}^\top \mathbf{A}$ entirely, bypassing this issue. It also leads to a clean formula for the projection matrix, and the final system can be solved efficiently via back substitution because $\mathbf{R}$ is upper triangular.

A.1 QR Factorization

Definition 5.2 (QR Factorization): Let $\mathbf{A}$ be an $m \times n$ matrix with linearly independent columns. The **QR factorization** of $\mathbf{A}$ is:

$$\mathbf{A} = \mathbf{Q}\mathbf{R} \quad (14)$$

where:

- $\mathbf{Q}$ is an $m \times n$ matrix with **orthonormal columns**: $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n$.
- $\mathbf{R}$ is an $n \times n$ **upper triangular** matrix with positive diagonal entries. Since $\mathbf{R}$ is upper triangular with nonzero diagonal entries, its determinant (equal to the product of the diagonal entries) is nonzero, so $\mathbf{R}$ is invertible.

The columns of $\mathbf{Q}$ form an orthonormal basis for $\text{Col}(\mathbf{A})$. The standard method for computing the QR factorization is the **Gram-Schmidt process**, which orthogonalizes the columns of $\mathbf{A}$ to produce $\mathbf{Q}$. From there, $\mathbf{R}$ can be obtained via the equation $\mathbf{R} = \mathbf{Q}^\top \mathbf{A}$.

A.2 Least Squares Solution via QR

Substituting $\mathbf{A} = \mathbf{Q}\mathbf{R}$ into the least squares formula $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$ yields the following:

$$\begin{aligned} \hat{\mathbf{x}} &= (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} \\ &= ((\mathbf{Q}\mathbf{R})^\top \mathbf{Q}\mathbf{R})^{-1} (\mathbf{Q}\mathbf{R})^\top \mathbf{b} \\ &= (\mathbf{R}^\top \mathbf{Q}^\top \mathbf{Q}\mathbf{R})^{-1} \mathbf{R}^\top \mathbf{Q}^\top \mathbf{b} \\ &= (\mathbf{R}^\top \mathbf{R})^{-1} \mathbf{R}^\top \mathbf{Q}^\top \mathbf{b} && \text{(since } \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n\text{)} \\ &= \mathbf{R}^{-1} (\mathbf{R}^\top)^{-1} \mathbf{R}^\top \mathbf{Q}^\top \mathbf{b} \\ &= \mathbf{R}^{-1} \mathbf{Q}^\top \mathbf{b} \end{aligned} \quad (15)$$

Therefore, the least squares solution can be written as:

$$\boxed{\hat{\mathbf{x}} = \mathbf{R}^{-1} \mathbf{Q}^\top \mathbf{b}} \quad (16)$$

In practice, this is computed in two steps:

1. Compute $\mathbf{c} = \mathbf{Q}^\top \mathbf{b}$.

2. Solve $\mathbf{R}\hat{\mathbf{x}} = \mathbf{c}$ by back substitution (since $\mathbf{R}$ is upper triangular, this is efficient).

Example 5.2 (Solving a least squares problem via QR):

This is the same system as in the linear regression example in Section 5.5.1, which we solved there using the normal equations.

Find the vector $\hat{\mathbf{x}}$ that minimizes $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}.$$

Step 1: Compute the QR factorization. We apply Gram-Schmidt to the columns $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$.

Normalize $\mathbf{a}_1$:

$$\|\mathbf{a}_1\| = \sqrt{3}, \quad \mathbf{q}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Subtract the projection of $\mathbf{a}_2$ onto $\mathbf{q}_1$ from $\mathbf{a}_2$:

$$\mathbf{a}_2 - (\mathbf{q}_1^\top \mathbf{a}_2) \mathbf{q}_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} - \sqrt{3} \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Normalize:

$$\left\| \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\| = \sqrt{2}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

So:

$$\mathbf{Q} = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}, \quad \mathbf{R} = \mathbf{Q}^\top \mathbf{A} = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}$$

Step 2: Compute $\mathbf{c} = \mathbf{Q}^\top \mathbf{b}$.

$$\mathbf{c} = \mathbf{Q}^\top \mathbf{b} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 9/\sqrt{3} \\ 4/\sqrt{2} \end{bmatrix}$$

Step 3: Solve $\mathbf{R}\hat{\mathbf{x}} = \mathbf{c}$ by back substitution.

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 9/\sqrt{3} \\ 4/\sqrt{2} \end{bmatrix}$$

From the second row: $\sqrt{2}\hat{x}_2 = \frac{4}{\sqrt{2}}$, so $\hat{x}_2 = \frac{4}{2} = 2$.

Substituting into the first row: $\sqrt{3}\hat{x}_1 + \sqrt{3}(2) = \frac{9}{\sqrt{3}} = 3\sqrt{3}$, so $\sqrt{3}\hat{x}_1 = 3\sqrt{3} - 2\sqrt{3} = \sqrt{3}$, giving $\hat{x}_1 = 1$.

Therefore, $\hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

A.3 Projection via QR

Since $\text{Col}(\mathbf{Q}) = \text{Col}(\mathbf{A})$ and $\mathbf{Q}$ has orthonormal columns, the projection of $\mathbf{b}$ onto $\text{Col}(\mathbf{A})$ takes a particularly simple form:

$$\mathbf{A}\hat{\mathbf{x}} = \mathbf{Q}\mathbf{R} \cdot \mathbf{R}^{-1}\mathbf{Q}^\top \mathbf{b} = \mathbf{Q}\mathbf{Q}^\top \mathbf{b} \quad (17)$$

Therefore, the projection matrix onto $\text{Col}(\mathbf{A})$ can also be written as:

$$\boxed{\mathbf{P} = \mathbf{Q}\mathbf{Q}^\top} \quad (18)$$

This is equivalent to $\mathbf{P} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$ and simpler to compute when $\mathbf{Q}$ is available.

A.4 Residual Norm via QR

We can derive a useful formula for the squared residual norm. Since $\mathbf{A}\hat{\mathbf{x}} = \mathbf{Q}\mathbf{Q}^\top \mathbf{b}$, the residual is:

$$\hat{\mathbf{r}} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}} = \mathbf{b} - \mathbf{Q}\mathbf{Q}^\top \mathbf{b} = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top) \mathbf{b}$$

We can use this result to decompose $\mathbf{b}$ into its projection and residual components:

$$\mathbf{b} = \underbrace{\mathbf{Q}\mathbf{Q}^\top \mathbf{b}}_{\in \text{Col}(\mathbf{Q})} + \underbrace{(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top) \mathbf{b}}_{\perp \text{Col}(\mathbf{Q})}$$

Since these two components are orthogonal, the Pythagorean theorem gives:

$$\|\mathbf{b}\|^2 = \|\mathbf{Q}\mathbf{Q}^\top \mathbf{b}\|^2 + \|(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top) \mathbf{b}\|^2$$

Rearranging and noting that $\|\mathbf{Q}\mathbf{Q}^\top \mathbf{b}\|^2 = (\mathbf{Q}^\top \mathbf{b})^\top (\mathbf{Q}^\top \mathbf{Q})(\mathbf{Q}^\top \mathbf{b}) = \|\mathbf{Q}^\top \mathbf{b}\|^2$:

$$\boxed{\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 = \|\mathbf{b}\|^2 - \|\mathbf{Q}^\top \mathbf{b}\|^2} \quad (19)$$

This gives a compact formula for the least squares error without needing to compute $\hat{\mathbf{x}}$ or the residual explicitly.